

S. K. Kiran

Portfolio | LinkedIn | GitHub
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PROFESSIONAL SUMMARY

Research Intern at NIT Calicut developing ROS2-based SLAM and autonomous navigation systems. Experienced in robotics, industrial automation, embedded systems, and computer vision with hands-on implementation of autonomous robotic and sensor-driven automation projects.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Embedded C, MATLAB
- **Robotics & Autonomous Systems:** ROS2, Gazebo, SLAM, Autonomous Navigation, Path Planning, Obstacle Avoidance, Sensor Fusion, Robot Kinematics
- **Embedded Systems:** ESP32, Arduino, GPIO, PWM, UART, RaspberryPi, Sensor Interfacing
- **Industrial Automation:** PLC (Delta), SCADA, HMI, WinCC, Ignition, Closed-Loop Control Systems
- **Computer Vision & AI:** OpenCV, Image Processing, Face Recognition, Object Detection, CNN-Based Classification
- **CAD & Engineering Tools:** SolidWorks, AutoCAD, MATLAB/Simulink, Power BI, Fusion360
- **Development Tools:** Linux (Ubuntu), Git, GitHub
- **Sensors & Actuators:** Ultrasonic, Gas, NPK, Load Cell, LDR, Servo, Stepper, BLDC, DC Motors, Relays, Solenoids

EDUCATION

Paavai Engineering College

Bachelor of Engineering – Mechatronics Engineering — CGPA: 7.54

Namakkal, Tamil Nadu

Oct 2023 – Present

N.V.K.S Higher Secondary School

Higher Secondary Certificate (HSC) — 74.8%

Attoor, Tamil Nadu

2023

RELEVANT COURSEWORK

Control Systems — Industrial Automation — Robotics — Embedded Systems — Sensors & Instrumentation — Machine Learning — Computer Vision — Mechatronics System Design

RESEARCH EXPERIENCE

National Institute of Technology Calicut (NIT Calicut)

Research Intern – Electrical Engineering Department

Kozhikode, Kerala

May 2026 – Present

- **SLAM Development:** Developing ROS2-based SLAM, localization, and autonomous navigation algorithms for wheeled mobile robots using real-time sensor integration and control systems.
- **Autonomous Navigation:** Implementing navigation and path-planning strategies for autonomous robot operation in dynamic and unknown environments.
- **Hardware Validation:** Testing and validating navigation algorithms on physical robotic platforms through real-world experimentation and hardware-in-the-loop evaluation.

INDUSTRIAL EXPERIENCE

SWOT Solar

Solar and Water Systems Intern

Udumalaipettai, Coimbatore

Jul 2025 – Aug 2025

- **PV System Analysis:** Evaluated photovoltaic system performance using MATLAB/Simulink and field measurements to improve energy utilization and system efficiency.
- **System Integration:** Performed component selection, load calculations, and subsystem integration for solar and water treatment applications.

- **Technical Documentation:** Prepared engineering reports and analyzed deployment constraints for renewable energy installations.

PROJECTS

- **Autonomous Waste Segregation System** GitHub
Designed and developed an Arduino-based automated waste segregation system integrating ultrasonic, gas, and load-cell sensors for multi-parameter waste classification. Led a team of 2 members. Implemented servo-driven sorting, real-time monitoring, and odor-control mechanisms for autonomous operation.
- **iAttend – Facial Recognition Attendance System** GitHub
Developed a real-time attendance management system using Python and OpenCV with facial recognition capabilities. Led a team of 4 members. Implemented automated attendance logging, database integration, and dashboard-based monitoring for managing 100+ student records.
- **Smart Fertilizer Application System**
Developed a smart agriculture system integrating CNN-based plant disease classification with NPK sensor feedback and automated fertilizer dispensing. Presented at ICATS'25 International Conference for its application in precision agriculture.
- **Solar Panel Alignment System Using Hydraulics**
Designed and fabricated a hydraulic solar tracking mechanism capable of adjusting panel orientation according to sunlight direction. Led a team of three members. Demonstrated improved solar exposure while maintaining a low-cost implementation suitable for educational and prototype applications.

CERTIFICATIONS

- OpenCV Bootcamp – OpenCV University (100%)
- Industrial Robotics & Automation – ABB / AMS
- Learn SCADA from Scratch – Udemy
- Operating Systems – Google / Coursera
- Supervised Machine Learning – DeepLearning.AI / Coursera

ACHIEVEMENTS

- Selected for presentation at ICATS'25 International Conference for the Smart Fertilizer Application System.
- Secured a Research Internship at NIT Calicut under the Summer Internship Programme 2026.
- Presented Solar Panel Alignment System Using Hydraulics at Techfinix'24 Technical Symposium.
- Achieved 100% score in OpenCV Bootcamp conducted by OpenCV University.

LANGUAGES

English (R/W/S) — Tamil (R/W/S) — Malayalam (R/W/S) — Hindi (R/W/S)